```
#include <stdio.h>
int main() {
    int n, i;
    int pid[20], at[20], bt[20], pr[20];
    int wt[20], tat[20], ct[20];
    int completed[20] = {0};
    int current_time = 0, completed_count = 0;
    int highest_priority, selected;
    float avg_wt = 0, avg_tat = 0;

    printf("1WA24CS217\n");

    printf("Enter number of processes: ");
    scanf("%d", &n);
    for(i = 0; i < n; i++) {
        pid[i] = i + 1;
        printf("P%d Arrival Time: ", i+1);
        scanf("%d", &at[i]);
        printf("P%d Burst Time: ", i+1);
        scanf("%d", &bt[i]);
        printf("P%d Priority: ", i+1);
        scanf("%d", &pr[i]);
    }
    while(completed_count < n) {
        highest_priority = 999;
        selected = -1;
        for(i = 0; i < n; i++) {
            if(at[i] <= current_time && completed[i] == 0) {
                if(pr[i] < highest_priority) {
                    highest_priority = pr[i];
```

```
                selected = i;
            }
        }
    }
    if(selected == -1) {
        current_time++;
    }
    else {
        ct[selected] = current_time + bt[selected];
        tat[selected] = ct[selected] - at[selected];
        wt[selected] = tat[selected] - bt[selected];
        current_time = ct[selected];
        completed[selected] = 1;
        completed_count++;
        avg_wt += wt[selected];
        avg_tat += tat[selected];
    }
}

printf("\nProcess\tAT\tBT\tPR\tCT\tTAT\tWT\n");

for(i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",
            pid[i], at[i], bt[i], pr[i], ct[i], tat[i], wt[i]);
}

avg_wt /= n;
avg_tat /= n;

printf("\nAverage Waiting Time = %.2f", avg_wt);
```

```
    printf("\nAverage Turnaround Time = %.2f\n", avg_tat);


    return 0;
}
```

OUTPUT:

```
1WA24CS217
Enter number of processes: 5
P1 Arrival Time: 0
P1 Burst Time: 3
P1 Priority: 5
P2 Arrival Time: 2
P2 Burst Time: 2
P2 Priority: 3
P3 Arrival Time: 3
P3 Burst Time: 5
P3 Priority: 2
P4 Arrival Time: 4
P4 Burst Time: 4
P4 Priority: 4
P5 Arrival Time: 6
P5 Burst Time: 1
P5 Priority: 1

Process AT      BT      PR      CT      TAT     WT
P1      0       3       5       3       3       0
P2      2       2       3       11      9       7
P3      3       5       2       8       5       0
P4      4       4       4       15      11      7
P5      6       1       1       9       3       2

Average Waiting Time = 3.20
Average Turnaround Time = 6.20
```